

Phineas Fritsch

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Financial actuarial mathematics senior at UCLA, June 2027, looking at product, actuarial and revenue cycle roles. Six of my projects are running; four are open to anyone with the link. The software is AI-assisted: I decide what gets built, split the work across several agents, and reject what comes back wrong.

EDUCATION

University of California, Los Angeles Expected June 2027

- DEGREE** B.S. Financial Actuarial Mathematics. Coursework also includes Probability (170E/170S), MGMT 1A/1B, 105, 109, ECON 1/11.
- ACTUARIAL** **MATH 174E, Mathematics of Finance**, is FM material and is behind me; Actuarial Models (178A, 179) enrolled for Fall 2026. Sitting Exam FM, Fall 2026; no exams passed to date.
- CASE COMP** 13th Annual Actuarial Case Competition (BAS), Winter 2025. Evaluated three commercial property insurance structures for a simulated Fortune 500 client against tail risk, fitting Lognormal, Gamma and Pareto to historical losses and running 10,000-iteration Monte Carlo simulations to compare TCOR and Risk-Adjusted TCOR.

EXPERIENCE

UCLA Library, Biomedical Library — Student Assistant II Mar 2026 – Present

- Nobody assigned any of this. I built three tools for the unit — the daily schedule, the shelf-face lookup, a rebuilt headcount page — plus an iOS app routing shelf-reading, shelving and ILL pulls. I wanted to be lazy about the parts that repeat daily.
- The daily staff schedule used to arrive as a spreadsheet every morning, so I put it on a phone screen. People at my desk open mine now instead of the file.
- After the library retired LibMaps, photographed and transcribed 453 shelf-end range labels myself (1,123 photographs), validated them into one dataset, and built a call-number-to-shelf lookup and pull-list route planner on it (shelfmark.phineasfritsch.com).
- Circulation, interlibrary-loan pull lists, shelf reading, headcounts, opening and closing. Promoted from Student Assistant I.

MKTaxSolutions — IT Intern Aug – Sep 2024

- Consolidated 400+ client profiles across three services; integrated CCH software with office scanning systems.

Vera A. Fritsch DMD — Operations Assistant Jun – Oct 2023

- My family's dental practice. Front-office clinic operations, patient data entry and financial record accuracy, on Dentrrix.

PROJECTS

Dibs — dibs.ge Rails 8 · PostgreSQL · Kamal · live

- Ranks UCLA general-education courses on grade history I obtained by filing four California Public Records Act requests: 176,290 rows, expanded to 1,544,628 individual grades after correcting an instructor fan-out that would have added 81,559 phantom ones.
- Rewrote the crawler from 657 + 480 requests an hour into a single ~700-request pass with a hard budget and a concurrency-1 lock: less load on UCLA's servers, and catalog staleness fell from 2,085 minutes to 3. Runs on a 2.5 GB server I budgeted by hand.

LEADERSHIP

UCLA Sailing Team — Treasurer, then Team Captain Jul 2024 – Jun 2026

- Ran the executive board and weekly meetings; coordinated travel, housing and rosters for biweekly regattas.

Theta Chi, Beta Alpha Chapter — House Manager, then Network Manager Jul 2025 – Present

- As Network Manager, built the chapter WordPress replacement: a static site a non-technical successor edits through a CMS, every legacy URL preserved. Running now; the domain cutover is mine to schedule.
- Managed a \$10,000 annual operating budget and daily property operations with alumni. Directed summer renovations (plumbing, structural repairs, a new game room) and used them in recruitment, contributing to the second-largest pledge class in chapter history.

SKILLS

- QUANTITATIVE** Monte Carlo simulation, loss distribution fitting, Excel modeling
- ENGINEERING** Ruby on Rails, PostgreSQL, TypeScript, Swift; Docker, Kamal, Cloudflare, GitHub Actions; Linux, Proxmox, networking. Runs 40 TB raw on a 2018 Dell Precision workstation, 96 GB DDR4, containerized since a dorm-room first attempt in 2023; a three-node R720 rack is bought and not yet deployed.
- LANGUAGES** English (native), German (B2/C1, accredited)